LUAD knowledge graph: evidence, targets and hypotheses

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1. Schema and evidence

The completed graph contains **27 nodes and 30 relationships**. It combines the original 15 LUAD candidates with contextual EGFR and TOP1 genes. Disease, phenotype, pathway, drug and variant nodes connect molecular observations to interpretable clinical questions. Contextual genes do not enter the candidate ranking. Approved symbols and stable gene identifiers support consistent merging (HUGO Gene Nomenclature Committee, n.d.).

Node type	n	Relationship type	n
disease	2	ASSOCIATED_WITH	17
drug	2	HAS_PHENOTYPE	4
gene	17	OCCURS_IN	1
pathway	3	PARTICIPATES_IN	6
phenotype	2	TARGETS	2
variant	1		

Cohort gene-disease links represent measured RNA, protein and mutation associations, not causal drivers (Kanaparthy, 2026a). Reactome supports matrix organization, lectin complement and EGFR pathway membership (Reactome, n.d.-a, n.d.-b, n.d.-c). FDA labels support drug mechanisms and indication-specific context. NCI supports disease-symptom links; HPO standardizes cough and dyspnea (Human Phenotype Ontology Consortium, n.d.-a, n.d.-b). ClinVar documents the external EGFR L858R variant, without asserting its presence in this cohort (National Center for Biotechnology Information, n.d.).

Every exported relationship includes its source URL and confidence rationale. **High:** Direct curated identifier/pathway assertion or regulatory drug mechanism/indication. **Low:** An indirect hypothesis without direct experimental support; not used as an asserted molecular edge. **Moderate:** Measured observational association in one cohort, or general clinical symptom description. These categories concern the stored edge, not causal certainty. GWAS p-values are unavailable and are not inferred from RNA p-values.

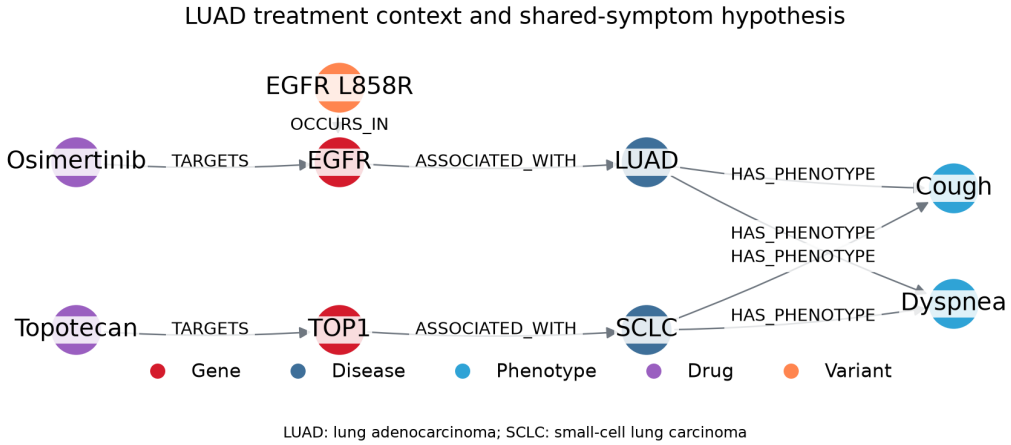


Figure 1. Focused drug, gene, disease and shared-phenotype paths from the actual graph, colored by node type. The full PNG accompanies this report; nodes.csv and edges.csv retain every node and relationship.

2. Targets, evidence fusion and communities

All 15 cohort genes have equal directed PageRank. Genes point toward disease and pathway nodes, while the incoming drug and variant edges enter contextual genes. The graph therefore provides no centrality discrimination among the original candidates. Their normalized centrality is set to the same neutral value, 0.5; this is a constant score component, not measured evidence.

Combined priority = 0.4C + 0.4M + 0.2T. M is the within-15 percentile of the original Module 2 multi-omics score; T is the within-15 percentile of max(p_sm, p_ab). C is normalized centrality, with 0.5 assigned when all values tie. The distinct raw p_sm and p_ab values remain visible below. These model-derived tractability priors are not calibrated probabilities of clinical success. Weights are declared teaching choices, rather than fitted or validated clinical parameters (Kanaparthy, 2026a, 2026b).

Rank	Gene	PageRank	Omics	p_sm	p_ab	Combined
1	ACTN2	0.01689	0.9922	0.2200	0.0004	0.7200
1	COL6A6	0.01689	0.9859	0.2804	0.0305	0.7200
3	SCN7A	0.01689	0.9824	0.6448	0.0004	0.6933
4	TNR	0.01689	0.9850	0.1404	0.2357	0.6533
5	ZNF536	0.01689	0.9876	0.0030	0.0004	0.5867

The leading candidates are **ACTN2, COL6A6, SCN7A, TNR, ZNF536**. ACTN2 and COL6A6 share the leading priority 0.7200; their display order reflects tie handling. The retained omics evidence and relative tractability drive these positions. They should not be attributed to a differentiated PageRank signal when centrality ties. Comparison: ACTN2: Module 2 rank 1, Module 3 rank 2; COL6A6: Module 2 rank 3, Module 3 rank 1; SCN7A: Module 2 rank 5, Module 3 rank 3. Spearman rank correlations with Modules 2 and 3 are 0.913 and 0.987, respectively.

ACTN2's leading rank reflects large decreases in both RNA and protein abundance, rather than evidence that inhibiting it would treat LUAD. COL6A6 connects to curated extracellular-matrix biology, supporting investigation of tissue composition and stromal remodeling. These cohort candidates require functional validation; EGFR provides separate established, genotype-dependent treatment context (Kanaparthy, 2026a; Reactome, n.d.-a; U.S. Food and Drug Administration, 2024).

ZNF536 illustrates the evidence limits: RNA log2 fold change -2.72 and protein change 2.71 have opposite signs; the protein estimate uses 15 valid pairs. A large change in abundance does not establish a therapeutic dependency.

Detected community and interpretation

Greedy modularity on the undirected graph found **5 communities**. The selected community contains COL11A1; COL6A6; Extracellular matrix organization; TNR; TNXB. Its genes are COL11A1, COL6A6, TNR, TNXB; its pathways are Extracellular matrix organization. Its matrix-related genes and pathway suggest extracellular-matrix remodeling or altered stromal composition as a LUAD-associated mechanism. This interpretation concerns tissue biology; mixed cell populations could produce the same abundance pattern (Reactome, n.d.-a). Community membership depends on annotation choices, graph direction removal and sparse connectivity. It does not establish an experimentally validated pathway or patient subtype.

3. Query results, hypothesis and limitations

The executed two-hop Drug → Gene → LUAD query returned: Osimertinib targets EGFR (FDA-approved in specified EGFR-mutant NSCLC settings; 2024 label). Approval refers to the source label's specified disease and molecular population, not every LUAD case (U.S. Food and Drug Administration, 2024). The shared-phenotype query returned 1 candidate row(s). Its actual topotecan route contains five nodes and four relationships; the last relationship is traversed in reverse.

Source node	Relationship	Destination node
Topotecan	TARGETS	TOP1
TOP1	ASSOCIATED_WITH	Small-cell lung carcinoma
Small-cell lung carcinoma	HAS_PHENOTYPE	Cough
Lung adenocarcinoma (LUAD)	HAS_PHENOTYPE	Cough

The topotecan label supports TOP1 inhibition and an SCLC treatment indication. TOP1's SCLC association therefore encodes therapeutic-target context, not an oncogenic driver claim. NCI documents cough and breathing difficulty in both clinical disease descriptions. A shared nonspecific symptom is weak evidence for drug transfer; this path does not show that topotecan treats LUAD (National Cancer Institute, n.d.-a, n.d.-b; U.S. Food and Drug Administration, 2019).

A testable hypothesis is that a molecularly defined LUAD subset could depend on TOP1 activity. Compare topotecan concentration-response curves in several LUAD models, nonmalignant lung controls and an SCLC comparator. Measure TOP1 engagement, DNA-damage markers and cell survival. Use an independent TOP1 perturbation and an appropriate rescue strategy to assess on-target dependence. Sensitivity, selectivity and reproducibility would be required before animal studies or any clinical interpretation; cough alone should not select treatment.

Uncertainty and checks

Cohort abundance associations may reflect tumor purity, stromal content or confounding. Selected pathway annotations are incomplete, and centrality changes with graph design. Within-list percentiles can amplify small original score differences. Tractability estimates come from a teaching classifier; COL6A6 and TNXB have truncated embedding representations (Kanaparthy, 2026b). Module 4 pocket results for SCN7A and MASP1 add structural context without proving ligand binding or LUAD dependence (Kanaparthy, 2026c). The optional patient layer is empty because Module 6 ages and disease labels are simulated. Module 5 CYP2C19 pharmacogenomic variants are not relabeled as LUAD driver mutations.

AI assisted identifier harmonization, source review, notebook coding and report drafting. Checks compared the original CSVs, official gene identifiers, source-specific biological claims, executed Neo4j query results and every reported path edge with the exported graph. Further progress requires independent LUAD cohorts, cell-type-resolved measurements, binding assays and functional perturbation. These checks support reproducibility, while the graph remains a hypothesis-generating synthesis.

References

Web resources were checked on November 2, 2026. FDA citations identify the specific label versions used; they are not described as the latest versions.

HUGO Gene Nomenclature Committee. (n.d.). *HGNC gene symbol reports and REST API*. https://www.genenames.org/

Human Phenotype Ontology Consortium. (n.d.-a). *Cough (HP:0012735)*. https://hpo.jax.org/browse/term/HP:0012735

Human Phenotype Ontology Consortium. (n.d.-b). *Dyspnea (HP:0002094)*. https://hpo.jax.org/browse/term/HP:0002094

Kanaparthy, A.. (2026a). *Module 2: LUAD integrated multi-omics results*. https://github.com/Ajaykanaparthy/biot6900/blob/main/module_2/data/luad_integrated_gene_level.csv

Kanaparthy, A.. (2026b). *Module 3: LUAD tractability predictions and target ranking*. https://github.com/Ajaykanaparthy/biot6900/blob/main/module_3/targets_luad_w3.csv

Kanaparthy, A.. (2026c). *Module 4: Verified structural pocket summary*. https://github.com/Ajaykanaparthy/biot6900/blob/main/module_4/results/verified_structure_summary.csv

National Cancer Institute. (n.d.-a). *Non-Small Cell Lung Cancer Treatment (PDQ) - Patient Version*. https://www.cancer.gov/types/lung/patient/non-small-cell-lung-treatment-pdq

National Cancer Institute. (n.d.-b). *Small Cell Lung Cancer Treatment (PDQ) - Patient Version*. https://www.cancer.gov/types/lung/patient/small-cell-lung-treatment-pdq

National Center for Biotechnology Information. (n.d.). *NM_005228.5(EGFR):c.2573T>G (p.Leu858Arg), Variation ID 16609*. https://www.ncbi.nlm.nih.gov/clinvar/variation/16609/

Reactome. (n.d.-a). *Extracellular matrix organization (R-HSA-1474244)*. https://reactome.org/content/detail/R-HSA-1474244

Reactome. (n.d.-b). *Lectin pathway of complement activation (R-HSA-166662)*. https://reactome.org/content/detail/R-HSA-166662

Reactome. (n.d.-c). *Signaling by EGFR (R-HSA-177929)*. https://reactome.org/content/detail/R-HSA-177929

U.S. Food and Drug Administration. (2019). *Topotecan injection: Prescribing information*. https://www.accessdata.fda.gov/drugsatfda_docs/label/2019/200582s003lbl.pdf

U.S. Food and Drug Administration. (2024). *Tagrisso (osimertinib) tablets: Prescribing information*. https://www.accessdata.fda.gov/drugsatfda_docs/label/2024/208065s033lbl.pdf